

CoTKR: Chain-of-Thought Enhanced Knowledge Rewriting for Complex Knowledge Graph Question Answering

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Abstract

Recent studies have explored the use of Large Language Models (LLMs) with Retrieval Augmented Generation (RAG) for Knowledge Graph Question Answering (KGQA). They typically require rewriting retrieved subgraphs into natural language formats comprehensible to LLMs. However, when tackling complex questions, the knowledge rewritten by existing methods may include irrelevant information, omit crucial details, or fail to align with the question’s semantics. To address them, we propose a novel rewriting method CoTKR, Chain-of-Thought Enhanced Knowledge Rewriting, for generating reasoning traces and corresponding knowledge in an interleaved manner, thereby mitigating the limitations of single-step knowledge rewriting. Additionally, to bridge the preference gap between the knowledge rewriter and the question answering (QA) model, we propose a training strategy PAQAF, Preference Alignment from Question Answering Feedback, for leveraging feedback from the QA model to further optimize the knowledge rewriter. We conduct experiments using various LLMs across several KGQA benchmarks. Experimental results demonstrate that, compared with previous knowledge rewriting methods, CoTKR generates the most beneficial knowledge representation for QA models, which significantly improves the performance of LLMs in KGQA.

1 Introduction

Large Language Models (LLMs) have achieved remarkable performance across various natural language processing tasks, marking a significant milestone (Sanh et al., 2022; Brown et al., 2020; Zhang et al., 2022). Despite their superior performance in zero-shot scenarios (Wei et al., 2022a; Kojima et al., 2022), they still encounter factual errors, known as “hallucinations” (Ji et al., 2023b), especially in knowledge-intensive tasks like question answering (QA) (Hu et al., 2023; Tan et al., 2023).

This issue arises due to the intrinsic limitations of LLMs, including factual inaccuracies and outdated knowledge. To address this challenge, a substantial of work (Ma et al., 2023a; Trivedi et al., 2023; Wu et al., 2023b) retrieves task-relevant knowledge from external sources as context, thereby enhancing the capabilities of LLMs in downstream tasks, known as Retrieval-Augmented Generation (RAG) (Lewis et al., 2020; Gao et al., 2023).

Recent work (Wu et al., 2023b; Baek et al., 2023; Sen et al., 2023; Wang et al., 2024) under the RAG paradigm explores the use of Knowledge Graphs (KGs) as an information source to enhance the capabilities of LLMs in Question Answering (QA). Unlike typical QA tasks, a key challenge in KGQA under this paradigm lies in transforming question-related subgraphs into natural language that LLMs can understand while preserving the structural information (Ko et al., 2024; Ding et al., 2024; Wu et al., 2023b). This process is referred to as Knowledge Rewriting (KR) in this study. As illustrated in Figure 1, this paper summarizes the commonly used knowledge rewriting methods in existing work. Most prior studies (Baek et al., 2023; Sen et al., 2023; Wang et al., 2023a) employ simple linear concatenation method (Triple), which concatenates the subject, relation, and object of each triple to form triple-form text. Additionally, considering that LLMs are pre-trained on text corpora and struggle with structured triple-form text, some efforts (Wu et al., 2023b; Bian et al., 2021; Chen et al., 2022) focus on converting triples into natural language through KG-to-Text. Furthermore, given that retrieved subgraphs often contain redundant information irrelevant to the question, other studies (Ko et al., 2024; Dernbach et al., 2024) aim to extract question-relevant knowledge from the triples to generate summary pertinent to the question.

Although these strategies are effective, they exhibit several limitations: (1) **Redundancy or omissions**. As illustrated in Figure 1, knowledge gen-

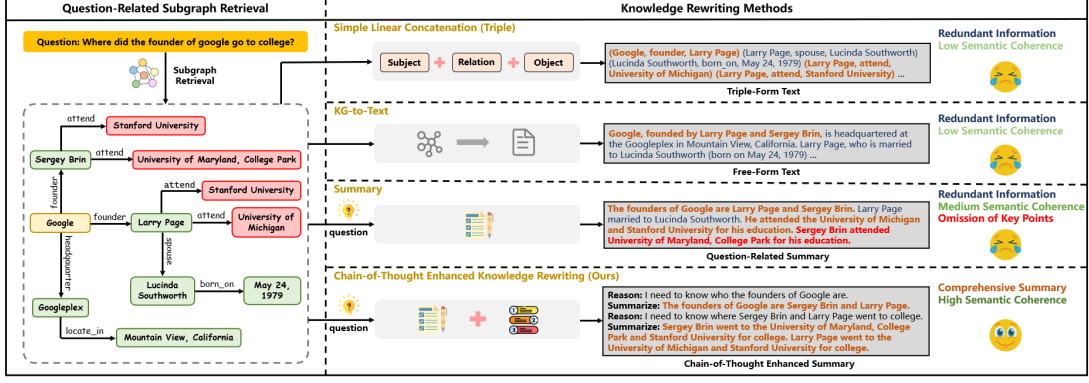


Figure 1: The commonly used knowledge rewriting methods in existing work.

erated by Triple and KG-to-Text are verbose, containing excessive irrelevant information. Summary provides a question-related summary but attempts to organize all relevant knowledge in one step. Given the extensive knowledge necessary to address complex questions, this method may not encapsulate all critical information, potentially resulting in the omission of key points. (2) **Semantic mismatch.** The three existing methods shown in Figure 1 ignore the semantics of the question and lack a logical organization that aligns with the question’s reasoning path.

To this end, we propose **Chain-of-Thought Enhanced Knowledge Rewriting**, CoTKR. Inspired by ReAct (Yao et al., 2023), the core of our method involves generating reasoning traces and corresponding knowledge in an interleaved manner. As shown in Figure 1, we alternate the following two operations: (1) **Reasoning**: decomposing the question to identify the knowledge required for inference; (2) **Summarization**: summarizing the relevant knowledge from retrieved triples, informed by the reasoning step’s output. By integrating Chain-of-Thought (CoT) (Wei et al., 2022b) with knowledge rewriting, CoTKR filters out irrelevant information and extracts question-related knowledge. Moreover, it generates a well-organized knowledge representation¹ semantically aligned with the question. Unlike traditional CoT applications in QA, our framework employs the knowledge rewriter to first summarize knowledge, which then serves as contextual information to enhance QA performance. This strategy offers superior robustness. Although the summary might be inaccurate, it still contributes valuable information, potentially leading to correct answers. However, ap-

¹In this paper, “knowledge representation” refers to the natural language form of question-related knowledge.

plying CoT to QA requires more precise reasoning chains, which are significantly affected by the error propagation (Wang et al., 2023a; Yao et al., 2023). To train knowledge rewriters based on LLMs, we design a training framework for CoTKR. In the first stage, inspired by previous work (Ma et al., 2023a; Wu et al., 2023b; Ko et al., 2024), we use knowledge representations generated by ChatGPT to guide the supervised fine-tuning of the knowledge rewriter, enabling it to initially master the capability of knowledge rewriting. In the second stage, we introduce **Preference Alignment** from **Question Answering Feedback (PAQAF)** to bridge the preference gap between the knowledge rewriter and the QA model. This method evaluates the quality of different knowledge representations based on the corresponding responses from the QA model. Subsequently, it constructs preference pairs, and fine-tunes LLMs through direct preference optimization (DPO) (Rafailov et al., 2023).

We conduct experiments on GrailQA (Gu et al., 2021) and GraphQuestions (Su et al., 2016), comparing commonly used knowledge rewriting methods in existing work. Contrary to previous findings (Dai et al., 2024; Baek et al., 2023), which suggest that LLMs perform better with knowledge in triple-form rather than in natural language, our findings demonstrate that LLMs can significantly benefit from knowledge represented in carefully crafted natural language. This indicates that our method could substantially enhance the performance of LLMs in KGQA.

The main contributions of this paper are:

- We propose CoTKR, a **Chain-of-Thought Enhanced Knowledge Rewriting** method to improve the quality of knowledge representation through the application of CoT. This method

generates reasoning traces and corresponding knowledge in an interleaved manner, thereby producing well-organized knowledge representations that are coherent with the question’s semantics.

- We propose a training strategy PAQAF, Preference Alignment from Question Answering Feedback, to bridge the preference gap between the knowledge rewriter and the QA model. It assesses the quality of different knowledge representations by evaluating corresponding responses from the QA model. Then, it constructs preference pairs and employs DPO to optimize the knowledge rewriter.
- We conduct experiments on two KGQA benchmarks. Compared with other knowledge rewriting methods, CoTKR can generate the most beneficial knowledge representation for the QA model and further enhance the performance of LLMs in KGQA. Additionally, considering privacy and cost issues, we evaluate the performance of open-source and closed-source LLMs as the foundational models for knowledge rewriting and QA.

2 Related Work

2.1 KG-Augmented LLMs for KGQA

To mitigate hallucination in LLMs, existing work (Wu et al., 2023b; Baek et al., 2023; Sen et al., 2023; Wang et al., 2024) attempts to enhance LLMs with KGs in the RAG paradigm. The naïve approach involves retrieving question-related triples from KGs as contextual information for QA (Baek et al., 2023; Sen et al., 2023). Although this method has proven effective, there remains ample room for improvement. Some studies (Wang et al., 2024, 2023a) integrate Chains-of-Thought (CoT) (Wei et al., 2022b) with RAG to tackle complex questions. Keqing(Wang et al., 2024) decomposes complex questions using predefined templates, retrieves candidate entities from KG, reasons through sub-questions, and ultimately generates answers with clear reasoning paths. KD-CoT (Wang et al., 2023a) validates and adjusts reasoning traces in CoT through interactions with external knowledge, thereby addressing issues of hallucinations and error propagation. Furthermore, alternative efforts (Wu et al., 2023b; Ko et al., 2024) address the limitations of LLMs in processing structured knowl-

edge and the noise in retrieved triples by post-processing these triples into natural language or summaries pertinent to the questions.

This paper focuses on optimizing the knowledge representation under the RAG paradigm for KGQA. Unlike previous work that transforms triples into the natural language in one step, we adopt CoT to summarize relevant knowledge step-by-step, ensuring comprehensiveness and semantic coherence in the generated knowledge.

2.2 Preference Alignment for LLMs on Question Answering

LLMs have the potential to generate content that contains gender discrimination, unethical elements, and racial biases, inconsistent with human values (Wu et al., 2023a; Ray, 2023). To address this issue, Preference Alignment (PA) (Ji et al., 2023a; Wang et al., 2023b) aims to fine-tune LLMs to align with human preferences. Existing QA work based on LLMs uses PA to bridge the gap between model preferences and those of humans or the QA tasks. KnowPAT (Zhang et al., 2023) trains LLMs on a knowledge preference set to align their knowledge biases with human preferences, selecting better factual knowledge as context. BGM (Ke et al., 2024) utilizes downstream task metrics as rewards to optimize the bridging model between retrievers and QA models. Rewrite-Retrieve-Read (Ma et al., 2023b) employs QA evaluation metrics as reward signals, fine-tuning the query rewriting module. EFSUM (Ko et al., 2024) constructs preference pairs sampled from LLMs and fine-tunes the knowledge summarizer using the Direct Preference Optimization (DPO) (Rafailov et al., 2023) algorithm.

Our study innovatively employs responses from the QA model to evaluate the quality of knowledge representations. We then construct preference pairs from these evaluations and optimize the knowledge rewriter using DPO.

3 Preliminaries

Knowledge Graph (KG) is a structured collection of triples in the form of (s, r, o) , where s, r, o represent the subject, the relation, and the object. This collection is denoted by $G = \{(s, r, o) \mid s, o \in E, r \in R\}$, where E represents the set of entities and R represents the set of relations.

Knowledge Graph Question Answering (KGQA) aims to answer natural language questions by utilizing a set of facts within KGs. Following previous

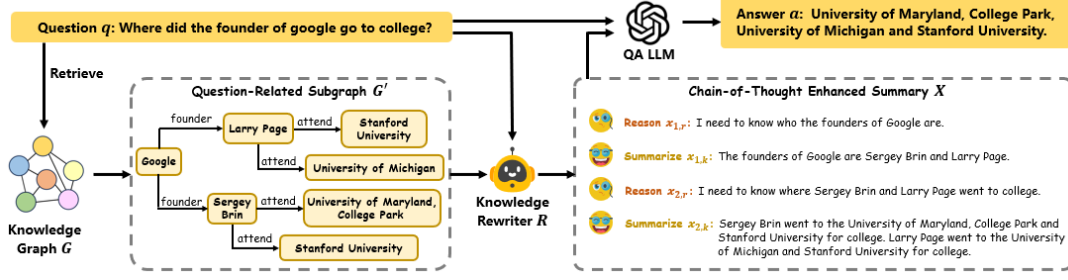


Figure 2: Illustration of our KGQA framework. CoTKR generates reasoning traces and corresponding knowledge in an interleaved manner.

work (Saxena et al., 2020; Jiang et al., 2023b), we assume that the subject entity of the question is given. Given a question q and a subject entity e , the objective is to generate a response a using the relevant facts in the KG G that accurately addresses the question.

Knowledge Rewriting (KR) for KGQA aims to transform question-related triples into natural language that can be consumed by LLMs. Given a question q and a subgraph $G' = \{(s, r, o) \mid s, o \in E, r \in R\}$ retrieved from KG G , the task is to generate a natural language sequence X that provides contextual information to answer the question.

4 Methods

4.1 Chain-of-Thought Enhanced Knowledge Rewriting

The architecture of our QA framework is depicted in Figure 2. Initially, our framework retrieves a question-related subgraph from the KG, which is subsequently transformed into contextual knowledge using CoTKR. This contextual knowledge, along with the question, prompts the QA model to generate an answer. The core of this framework is the knowledge rewriter. Briefly, it alternatively conducts the following two operations: **Reasoning**: decomposing the question and generating a reasoning trace based on generated knowledge representation and pointing out the specific knowledge needed for the current step; **Summarization**: summarizing the relevant knowledge based on the current reasoning trace from the subgraph.

Assume we have the reasoning traces at step $t - 1$ as $x_{t-1,r}$ and the summarized knowledge at step $t - 1$ as $x_{t-1,k}$. The corresponding knowledge representation, i.e., X_{t-1} is represented as:

$$X_{t-1} = [x_{1,r}, x_{1,k}, \dots, x_{t-1,r}, x_{t-1,k}]. \quad (1)$$

For knowledge rewriting at step t , given the ques-

tion q , the subgraph G' , and the previously generated content X_{t-1} , the knowledge rewriter R first generates the reasoning trace $x_{t,r}$:

$$x_{t,r} = R(q, G', X_{t-1}) \quad (2)$$

Subsequently, based on the question q , the subgraph G' , the previously generated content X_{t-1} , and the reasoning trace at step t $x_{t,r}$, CoTKR summarizes relevant knowledge $x_{t,k}$:

$$x_{t,k} = R(q, G', X_{t-1}, x_{t,r}) \quad (3)$$

$x_{t,r}$ and $x_{t,k}$ are attached to X_{t-1} for the knowledge representation at step t . Note in step 1, X_0 is initialized to None.

4.2 Training Framework for CoTKR

Fig. 3 illustrates the training framework of CoTKR.

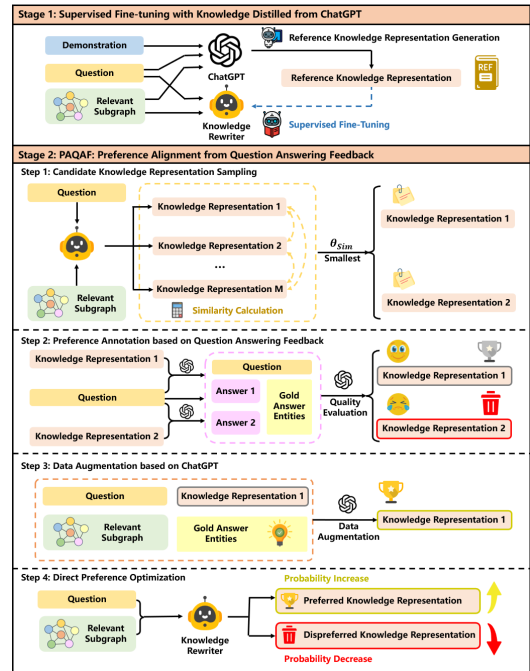


Figure 3: Our training framework for CoTKR.

4.2.1 Supervised Fine-tuning with Knowledge Distilled from ChatGPT

This stage enables open-source LLMs to initially acquire the knowledge rewriting capability through supervised fine-tuning. This primarily comprises two steps: reference knowledge representation generation and supervised fine-tuning.

Reference Knowledge Representation Generation. Inspired by previous work (Ma et al., 2023a; Wu et al., 2023b; Ko et al., 2024), we employ ChatGPT as the data generator to construct training corpora. We verbalize the question-related subgraph, G' , through simple linear concatenation and combine it with the question, q , to form the input prompt x . Subsequently, ChatGPT generates the reference knowledge representation k based on several examples (i.e., demonstrations) and the provided input x . Finally, we construct the training dataset $D_T = \{(x_1, k_1), (x_2, k_2), \dots, (x_T, k_T)\}$.

Supervised Fine-tuning. For every pair of input and output (x_i, k_i) in the training dataset D_T , our knowledge rewriter R_θ is trained to generate k_i based on x_i using the following objective:

$$L_{SFT} = -\frac{1}{T} \sum_{i=1}^T \log p_\theta(k_i|x_i) \quad (4)$$

where θ represents the parameters of the knowledge rewriter R_θ and $p_\theta(k_i|x_i)$ signifies the probability that R_θ generates k_i , given the input x_i .

4.2.2 Preference Alignment from Question Answering Feedback (PAQAF)

In this stage, Preference Alignment (PA) is employed to bridge the preference gap between the knowledge rewriter and the QA model. This stage includes four steps: candidate knowledge representation sampling, preference annotation based on QA feedback, data augmentation based on ChatGPT, and direct preference optimization (DPO).

Candidate Knowledge Representation Sampling. We input the question q and the corresponding subgraph G' , then sample M candidate knowledge representations, $k_1, k_2, \dots, k_M$, from the knowledge rewriter R_θ .

Preference Annotation based on Question Answering Feedback. Among the candidate knowledge representations, we select the two, k_1 and k_2 , with the greatest semantic difference (i.e., the lowest similarity) to facilitate faster convergence during training. Utilizing standard evaluation methods for assessing these representations is suboptimal, as they fail to align with the preferences

of QA models. Inspired by the findings in previous work (Wu et al., 2023b; Ko et al., 2024; Zhang et al., 2023), we posit that better knowledge representations generally lead to better performance on QA. Consequently, we adopt k_1 and k_2 as contextual knowledge, prompting the QA model Q to answer the question q , generating answers a_1 and a_2 , respectively. Subsequently, we prompt ChatGPT to assess the quality of a_1 and a_2 from the perspectives of accuracy and relevance. This evaluation aims to identify the preferred knowledge representation k^+ and the dispreferred knowledge representation k^- . Details of the evaluation prompt are provided in Appendix B.3.

Data Augmentation based on ChatGPT. ChatGPT is able to produce higher-quality knowledge representations, compared with open-source LLMs. Therefore, in order to improve the quality of preferred knowledge representation and enhance the diversity of the training data, we employ ChatGPT to paraphrase k^+ . In addition to the question q , the retrieved subgraph G' , and the preferred knowledge representation k^+ , we also provide the answer entity e . This allows ChatGPT to organize relevant knowledge around e , ensuring that the rewritten knowledge covers key evidence. We concatenate the question q and the textualized subgraph G' using a prompt template as the input x , and use the paraphrased knowledge representation k^{++} and k^- as the preferred pair. Finally, we construct the preference dataset $P_N = (x_1, k_1^{++}, k_1^-), (x_2, k_2^{++}, k_2^-), \dots, (x_N, k_N^{++}, k_N^-)$. The prompt for knowledge augmentation is in Appendix B.4.

Direct Preference Optimization. We employ Direct Preference Optimization (DPO) on our knowledge rewriter, R_θ , to develop a preference-tuned version, R_{θ^*} . It minimizes the following objective:

$$L_{DPO}(\theta^*; \theta) = -\frac{1}{N} \sum_{i=1}^N \log \sigma[r(x_i, k_i^{++}) - r(x_i, k_i^-)] \quad (5)$$

$$r(x_i, k_i) = \frac{p_{\theta^*}(k_i|x_i)}{p_\theta(k_i|x_i)} \quad (6)$$

Considering the varying preferences of different QA models, CoTKR is specifically trained for each QA model. Through the two stages of training, CoTKR tends to generate more favorable knowledge representation k^{++} for each QA model, while avoiding unhelpful knowledge representation k^- .

5 Experiments

5.1 Datasets

GrailQA (Gu et al., 2021) is a challenging, large-scale multi-hop KGQA benchmark that consists of 64,331 questions (44,337 train, 6,763 dev, 13,231 test). The training and dev sets provide annotated SPARQL query and answer entities, while the test set comprises only the questions. For evaluation convenience, the dev set is used for testing.

GraphQuestions (Su et al., 2016) is a characteristic-rich dataset for factoid question answering based on Freebase. It comprises 5,166 questions (2,771 train, 2,395 test). For each question, the dataset provides corresponding SPARQL query and answer entities.

5.2 Large Language Models

In this experiment, Llama-2 (7B) (Touvron et al., 2023b), Llama-3 (8B) (AI@Meta, 2024), and ChatGPT² are employed for knowledge rewriting, while ChatGPT and Mistral (7B) (Jiang et al., 2023a) are used for QA tasks. The details of these LLMs are provided in Appendix A.3.

5.3 Baselines

We compare **CoTKR** (without PAQAF) and **CoTKR+PAQAF** (**CoTKR+PA** for shortness) with other knowledge rewriting methods in KGQA: **Simple linear concatenation (Triple)** (Baek et al., 2023; Sen et al., 2023) concatenates the subject, predicate, and object of a triple to generate triple-form text. This method does not require additional models for knowledge rewriting.

KG-to-Text (Wu et al., 2023b) transforms facts into the free-form text for each relation path with a KG-to-Text model, addressing the limitations of LLMs in understanding structured triple-form text. **Summary** (Ko et al., 2024) converts triples into a question-relevant summary, alleviating the issue of redundant contextual knowledge.

To ensure a fair comparison, we employ the same corpus generation method for both the baselines and our method. All baselines undergo supervised fine-tuning without preference alignment.

5.4 Retrieval Methods

The retrieval module is not the focus of our research. Therefore, we adopt three commonly used retrieval methods. For detailed implementation, please refer to Appendix A.4.

2-Hop. We retain 30 triples from the 2-hop subgraph of the head entity, prioritizing those with higher semantic similarity to the question.

BM25. We follow the processing method in DecAF (Yu et al., 2023), simply linearizing the 1-hop subgraph of the topic entity as the article. We take the top 30 triples corresponding to the candidate documents as the retrieval results.

Ground Truth Subgraph (GS). We modify the SPARQL queries from the datasets to obtain the ground truth subgraphs. These subgraphs represent the results of an ideal retriever.

5.5 Evaluation Metrics

Following previous work on generative KGQA (Wu et al., 2023b; Baek et al., 2023; Ko et al., 2024), we adopt **Accuracy (Acc)** as one of our evaluation metrics. It measures whether the generated answer includes at least one correct answer entity. To provide a more comprehensive evaluation, we also incorporate **Recall** which measures the proportion of correct entities present in the model’s response, and **Exact Match (EM)** which evaluates whether the response contains all the answer entities.

5.6 Main Results

To comprehensively evaluate various knowledge rewriting methods, we employ the widely used 2-Hop retrieval method. Table 1 presents the overall results. We observe that: (1) **Our method outperforms the baselines across most evaluation metrics and LLMs, confirming the effectiveness of our knowledge rewriting strategy.** This also demonstrates the broad practical applicability of CoTKR, effective for both open-source LLMs requiring fine-tuning and closed-source LLMs using ICL. Integrating question-related knowledge significantly improves QA performance compared with direct question answering, underscoring the efficacy of the RAG paradigm in KGQA. KG-to-Text exhibits the weakest performance, indicating that mere conversion of triples into text may result in loss of information inherent in the subgraph. Summary outperforms KG-to-Text but generally lags behind CoTKR/CoTKR+PA, suggesting that filtering out irrelevant knowledge is effective but not adequate. (2) **CoTKR+PA matches or even surpasses the performance of ChatGPT as the knowledge rewriter, proving the effectiveness of our training framework and the preference alignment.** CoTKR+PA outperforms CoTKR, indicating that preference alignment can bridge the

²<https://api.openai.com/>

KR LLMs	Methods	GrailQA			GraphQuestions		
		Acc	Recall	EM	Acc	Recall	EM
ChatGPT as QA model							
None	No Knowledge	28.91	22.81	20.14	35.87	25.76	22.09
	Triple	57.76	49.67	44.73	55.03	46.65	41.63
Llama-2	KG-to-Text	54.75	47.35	42.44	49.73	40.00	33.74
	Summary	58.14	51.38	46.38	52.94	44.70	38.41
	CoTKR	58.64	52.33	47.88	51.36	45.20	39.96
	CoTKR+PA	59.25	53.52	49.64	56.78	47.99	42.46
Llama-3	KG-to-Text	55.76	48.41	43.90	52.40	45.06	39.83
	Summary	57.55	51.06	46.80	54.95	46.86	40.75
	CoTKR	58.33	52.55	48.65	53.19	47.23	43.17
	CoTKR+PA	61.51	56.08	52.67	56.37	49.31	45.26
ChatGPT	KG-to-Text	56.32	49.05	44.73	53.53	45.59	41.17
	Summary	58.54	51.81	47.29	55.62	48.93	44.97
	CoTKR	59.87	53.19	49.02	54.28	48.18	44.68
Mistral as QA model							
None	No Knowledge	29.44	23.13	20.30	38.20	26.92	22.13
	Triple	54.47	47.78	43.25	51.32	45.97	41.67
Llama-2	KG-to-Text	49.49	42.91	38.41	44.59	37.98	32.82
	Summary	54.10	47.79	43.15	49.85	42.33	36.45
	CoTKR	56.75	51.10	46.71	50.19	43.73	38.54
	CoTKR+PA	58.15	52.98	49.13	55.07	47.02	41.71
Llama-3	KG-to-Text	50.64	44.32	40.13	49.06	43.04	38.25
	Summary	53.84	47.71	43.49	52.03	44.30	38.50
	CoTKR	56.47	51.33	47.36	52.65	46.48	42.21
	CoTKR+PA	59.31	54.13	50.24	54.82	47.76	43.09
ChatGPT	KG-to-Text	51.04	44.87	40.97	49.14	43.04	38.83
	Summary	54.44	48.16	43.97	52.28	47.10	43.30
	CoTKR	57.28	51.14	47.09	52.82	47.13	43.55

Table 1: The overall results of CoTKR and the baselines on GrailQA and GraphQuestions using 2-Hop as retrieval method. For each combination of a knowledge rewriter (KR) LLM and a QA model, the best and second-best results are highlighted in bold and underlined, respectively.

preference gap between the knowledge rewriter and the QA model, thereby enhancing the quality of knowledge representation. (3) **A well-crafted knowledge representation is crucial for LLM used in KGQA.** Although Triple does not require an additional knowledge rewriting module, it provides a strong baseline and, in some cases, outperforms KG-to-Text and Summary. Conversely, CoTKR+PA consistently surpasses Triple. This indicates that Triple is simple yet effective and explains its widespread use in existing work. On the other hand, it demonstrates that a carefully designed knowledge representation can effectively enhance the performance of KGQA.

5.7 Impact of Retrieval Methods

To investigate the impact of retrieval methods, we select Llama-3 as the knowledge rewriter and ChatGPT as the QA model. According to the results shown in Figure 4, we have the following observations: (1) **2-Hop retrieval method may be insufficient for more challenging questions, but it is suitable for simpler ones.** Both BM25 and 2-Hop perform similarly on GrailQA, but 2-Hop shows a significant advantage over BM25 on GraphQuestions. This is likely because GrailQA is a more complex benchmark with a larger question-related subgraph, making 2-hop subgraphs often inadequate.

Conversely, for GraphQuestions, a 2-hop subgraph usually provides precise context for most questions. (2) **The design of a high-quality retriever remains an open problem.** GS significantly outperforms BM25 and 2-Hop, indicating that retrieval noise substantially affects KGQA performance. (3) **CoTKR consistently outperforms all baselines across various retrieval methods, demonstrating its robustness and practicality.**

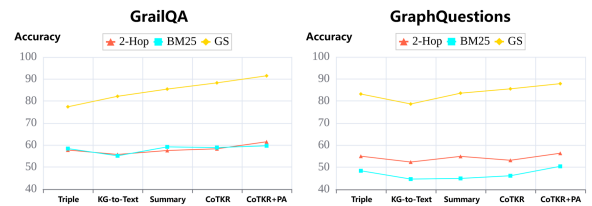


Figure 4: KGQA results using different knowledge rewriters and different retrieval methods.

5.8 Comparison with Triple Method

Several studies (Dai et al., 2024; Baek et al., 2023) suggest that LLMs can better comprehend triple-form text compared with natural language. However, our results show the contrary. Therefore, we delve deeper into this issue by comparing knowledge rewriting methods that use triple-form text as input (i.e., KG-to-Text, Summary, CoTKR,

CoTKR+PA) with Triple. We use Accuracy as the criterion to evaluate the correctness of responses. For each method, we consider three scenarios: (1) **Incorrect→Correct**: Triple provides a wrong answer, but the comparative method answers correctly. (2) **Correct→Incorrect**: Triple answers correctly, but the comparative method answers incorrectly. (3) **No change**: both Triple and the comparative method answer correctly or incorrectly. We adopt Llama-3 as the knowledge rewriter and ChatGPT as the QA model, with 2-Hop as the retrieval method. Then we calculate the proportions of three distinct cases within GrailQA. From the results shown in Figure 5, we draw the following conclusions: (1) **KG-to-Text and Summary have a predominantly negative impact, partially validating the conclusions of prior studies**. Triple provides a strong baseline, and the adoption of KG-to-Text and Summary leads to more incorrect answers. This indicates that LLMs can understand triple-form text effectively, and using simple knowledge rewriting methods leads to loss of information. (2) **Well-designed knowledge representations substantially benefit the question-answering model**. The knowledge representations rewritten by CoTKR/CoTKR+PA generally enhance the QA model’s performance. This reflects that the suboptimal knowledge representations in previous work are key contributors to performance degradation. Our method generates comprehensive and semantically coherent knowledge representations, thereby improving the efficacy of KGQA.

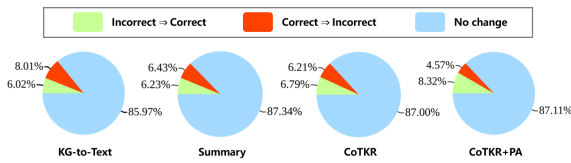


Figure 5: The comparative results on GrailQA. We use 2-Hop as the retrieval method.

5.9 Effectiveness of Data Augmentation

To evaluate the effectiveness of data augmentation, we compare the performance of three variants: CoTKR, CoTKR+PA, and CoTKR+PA* (using supervised fine-tuning and preference alignment without data augmentation). The experimental results are shown in Table 2, from which we can draw two conclusions: (1) **CoTKR+PA* generally outperforms CoTKR+PA, indicating that PAQAF does not solely rely on data augmentation based**

on ChatGPT. (2) CoTKR+PA performs best in most scenarios, proving that data augmentation enhances the preference alignment.

KR LLMs	Methods	ChatGPT			Mistral		
		Acc	Recall	EM	Acc	Recall	EM
Llama-2	CoTKR	51.36	45.20	39.96	50.19	43.73	38.54
	CoTKR+PA*	54.57	47.40	42.88	52.99	45.69	41.09
	CoTKR+PA	56.78	47.99	42.46	55.07	47.02	41.71
Llama-3	CoTKR	53.19	47.23	43.17	52.65	46.48	42.21
	CoTKR+PA*	54.70	48.79	44.72	53.24	47.64	43.63
	CoTKR+PA	56.37	49.31	45.26	54.82	47.76	43.09

Table 2: The ablation results of data augmentation on GraphQuestions. We use 2-Hop as the retrieval method.

5.10 Case Study

In this section, we compare Summary with CoTKR through an example. (Please refer to Appendix C for the full comparison result.) As illustrated in Table 3, Summary overlooks the key knowledge (i.e., Stephen Jay Gould). In contrast, CoTKR highlights the most relevant knowledge in the first rewriting step (i.e., Knowledge 1). Furthermore, after preference alignment, CoTKR+PA can avoid the unnecessary reasoning step (i.e., Reason 2), generating comprehensive and semantic coherent contextual knowledge for QA.

Question: what is the central focus of the radio program episode <i>talk of the nation: memes</i> ?	
Answer: <i>Stephen Jay Gould</i>	
KR Methods	KR Results
Summary	The central focus of the radio program episode "Talk of the Nation: Memes" is discussing topics related to memes and their impact on society. Reason: I need to know the subject or focus of the radio program episode "Talk of the Nation: Memes".
CoTKR	Summarize: The subject of the radio program episode "Talk of the Nation: Memes" is <i>Stephen Jay Gould</i> . Reason: I need to know more about Stephen Jay Gould. Summarize: <i>Stephen Jay Gould</i> was a person with nationality ... Reason: I need to know the subject or focus of the radio program episode "Talk of the Nation: Memes".
CoTKR+PA	Summarize: The subject of the radio program episode "Talk of the Nation: Memes" is <i>Stephen Jay Gould</i> .

Table 3: An example of knowledge rewriting results for different methods.

6 Conclusion

In this paper, we propose Chain-of-Thought Enhanced Knowledge Rewriting, CoTKR, for higher quality knowledge representation of triples in KG augmented QA. To bridge the preference gap between the knowledge rewriter and the QA model, we propose Preference Alignment from Question Answering Feedback, PAQAF. Experimental results demonstrate that, compared with existing knowledge rewriting methods, CoTKR can generate the most beneficial knowledge representation for QA models. In future work, we will go beyond KGQA to explore knowledge representations for other kinds of structured data for RAG.

7 Limitations

We acknowledge the limitations of this work. (1) This study is limited to KGQA and does not explore broader application scenarios. Therefore, we did not design experiments to explore whether CoTKR is effective for all or most RAG scenarios. In future work, we aim to expand the range of data sources. We intend to design a knowledge rewriting method that can be applied to not only KGs but also tables, textual data, and other formats. This enhancement will allow the QA framework to access knowledge from a wider range of sources, thus improving its practicality. Furthermore, we plan to investigate a knowledge representation beneficial for various downstream tasks, such as fact verification and dialogue generation. (2) The training framework for CoTKR depends on the powerful capabilities of closed-source LLMs. However, these models have inherent limitations, and the training data they generate contains noise, which constrains the performance ceiling of CoTKR.

8 Ethical Considerations

We explore optimizing knowledge representations for KGQA on public benchmarks, avoiding any potential harm to any individuals or groups. To promote transparency and facilitate replication of our research, we provide the technical details necessary for reproducing our results and release both the source code and the collected data. Our code and data are available for academic research, commercial use, and other applications.

It is important to acknowledge the potential risks and ethical considerations associated with LLMs. In this study, we construct the training data using ChatGPT and implement our knowledge rewriters based on LLMs. Due to the inherent limitations of LLMs, including factual inaccuracies, racial discrimination, and gender bias, our knowledge rewriters might generate incorrect content or inadvertently reflect prevalent societal biases.

References

AI@Meta. 2024. [Llama 3 model card](#).
Jinheon Baek, Alham Fikri Aji, and Amir Saffari. 2023. [Knowledge-augmented language model prompting for zero-shot knowledge graph question answering](#). In *Proceedings of the 1st Workshop on Natural Language Reasoning and Structured Explanations (NLRSE)*, pages 78–106, Toronto, Canada. Association for Computational Linguistics.

Ning Bian, Xianpei Han, Bo Chen, and Le Sun. 2021. Benchmarking knowledge-enhanced commonsense question answering via knowledge-to-text transformation. In *AAAI*, pages 12574–12582. AAAI Press.
Kurt D. Bollacker, Robert P. Cook, and Patrick Tufts. 2007. Freebase: A shared database of structured general human knowledge. In *AAAI*, pages 1962–1963. AAAI Press.
Kurt D. Bollacker, Colin Evans, Praveen K. Paritosh, Tim Sturge, and Jamie Taylor. 2008. Freebase: a collaboratively created graph database for structuring human knowledge. In *SIGMOD Conference*, pages 1247–1250. ACM.
Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. 2020. Language models are few-shot learners. In *NeurIPS*.
Zhuo Chen, Yufeng Huang, Jiaoyan Chen, Yuxia Geng, Yin Fang, Jeff Z. Pan, Ningyu Zhang, and Wen Zhang. 2022. Lako: Knowledge-driven visual question answering via late knowledge-to-text injection. In *IJCKG*, pages 20–29. ACM.
Xinbang Dai, Yuncheng Hua, Tongtong Wu, Yang Sheng, and Guilin Qi. 2024. Counter-intuitive: Large language models can better understand knowledge graphs than we thought. *CoRR*, abs/2402.11541.
Stefan Dernbach, Khushbu Agarwal, Alejandro Zuniga, Michael Henry, and Sutanay Choudhury. 2024. Glam: Fine-tuning large language models for domain knowledge graph alignment via neighborhood partitioning and generative subgraph encoding. In *AAAI Spring Symposia*, pages 82–89. AAAI Press.
Wentao Ding, Jinmao Li, Liangchuan Luo, and Yuzhong Qu. 2024. Enhancing complex question answering over knowledge graphs through evidence pattern retrieval. In *WWW*, pages 2106–2115. ACM.
Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, Qianyu Guo, Meng Wang, and Haofen Wang. 2023. Retrieval-augmented generation for large language models: A survey. *CoRR*, abs/2312.10997.
Yu Gu, Sue Kase, Michelle Vanni, Brian M. Sadler, Percy Liang, Xifeng Yan, and Yu Su. 2021. Beyond I.I.D.: three levels of generalization for question answering on knowledge bases. In *WWW*, pages 3477–3488. ACM / IW3C2.
Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and

719	Weizhu Chen. 2022. Lora: Low-rank adaptation of large language models. In <i>ICLR</i> . OpenReview.net.	
720		
721	Nan Hu, Yike Wu, Guilin Qi, Dehai Min, Jiaoyan Chen, Jeff Z. Pan, and Zafar Ali. 2023. An empirical study of pre-trained language models in simple knowledge graph question answering. <i>World Wide Web (WWW)</i> , 26(5):2855–2886.	
722		
723		
724		
725		
726	Jiaming Ji, Tianyi Qiu, Boyuan Chen, Borong Zhang, Hantao Lou, Kaile Wang, Yawen Duan, Zhonghao He, Jiayi Zhou, Zhaowei Zhang, Fanzhi Zeng, Kwan Yee Ng, Juntao Dai, Xuehai Pan, Aidan O’Gara, Yingshan Lei, Hua Xu, Brian Tse, Jie Fu, Stephen McAleer, Yaodong Yang, Yizhou Wang, Song-Chun Zhu, Yike Guo, and Wen Gao. 2023a. AI alignment: A comprehensive survey. <i>CoRR</i> , abs/2310.19852.	
727		
728		
729		
730		
731		
732		
733		
734		
735	Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Yejin Bang, Andrea Madotto, and Pascale Fung. 2023b. Survey of hallucination in natural language generation. <i>ACM Comput. Surv.</i> , 55(12):248:1–248:38.	
736		
737		
738		
739		
740	Albert Q Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. 2023a. Mistral 7b. <i>arXiv preprint arXiv:2310.06825</i> .	
741		
742		
743		
744		
745	Jinhao Jiang, Kun Zhou, Xin Zhao, and Ji-Rong Wen. 2023b. Unikgqa: Unified retrieval and reasoning for solving multi-hop question answering over knowledge graph. In <i>ICLR</i> . OpenReview.net.	
746		
747		
748		
749	Zixuan Ke, Weize Kong, Cheng Li, Mingyang Zhang, Qiaozhu Mei, and Michael Bendersky. 2024. Bridging the preference gap between retrievers and llms. <i>CoRR</i> , abs/2401.06954.	
750		
751		
752		
753	SungHo Ko, Hyunjin Cho, Hyungjoo Chae, Jinyoung Yeo, and Dongha Lee. 2024. Evidence-focused fact summarization for knowledge-augmented zero-shot question answering. <i>CoRR</i> , abs/2403.02966.	
754		
755		
756		
757	Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. 2022. Large language models are zero-shot reasoners. In <i>NeurIPS</i> .	
758		
759		
760	Patrick S. H. Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich K�ttler, Mike Lewis, Wen-tau Yih, Tim Rockt�schel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In <i>NeurIPS</i> .	
761		
762		
763		
764		
765		
766	Jimmy Lin, Xueguang Ma, Sheng-Chieh Lin, Jheng-Hong Yang, Ronak Pradeep, and Rodrigo Frassetto Nogueira. 2021. Pyserini: A python toolkit for reproducible information retrieval research with sparse and dense representations. In <i>SIGIR</i> , pages 2356–2362. ACM.	
767		
768		
769		
770		
771		
	Xinbei Ma, Yeyun Gong, Pengcheng He, Hai Zhao, and Nan Duan. 2023a. Query rewriting in retrieval-augmented large language models. In <i>EMNLP</i> , pages 5303–5315. Association for Computational Linguistics.	772 773 774 775 776
	Xinbei Ma, Yeyun Gong, Pengcheng He, Hai Zhao, and Nan Duan. 2023b. Query rewriting in retrieval-augmented large language models . In <i>Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing</i> , pages 5303–5315, Singapore. Association for Computational Linguistics.	777 778 779 780 781 782
	Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D. Manning, Stefano Ermon, and Chelsea Finn. 2023. Direct preference optimization: Your language model is secretly a reward model. In <i>NeurIPS</i> .	783 784 785 786
	Partha Pratim Ray. 2023. Chatgpt: A comprehensive review on background, applications, key challenges, bias, ethics, limitations and future scope. <i>internet of things and cyber-physical systems</i> , 3, 121–154. <i>URL</i> https://doi.org/10.1016/j.iotcps.3 .	787 788 789 790 791
	Stephen E. Robertson and Hugo Zaragoza. 2009. The probabilistic relevance framework: BM25 and beyond. <i>Found. Trends Inf. Retr.</i> , 3(4):333–389.	792 793 794
	Victor Sanh, Albert Webson, Colin Raffel, Stephen H. Bach, Lintang Sutawika, Zaid Alyafeai, Antoine Chaffin, Arnaud Stiegler, Arun Raja, Manan Dey, M Saiful Bari, Canwen Xu, Urmish Thakker, Shanya Sharma Sharma, Eliza Szczechla, Taewoon Kim, Gunjan Chhablani, Nihal V. Nayak, Debajyoti Datta, Jonathan Chang, Mike Tian-Jian Jiang, Han Wang, Matteo Manica, Sheng Shen, Zheng Xin Yong, Harshit Pandey, Rachel Bawden, Thomas Wang, Trishala Neeraj, Jos Rozen, Abheesht Sharma, Andrea Santilli, Thibault F�vry, Jason Alan Fries, Ryan Teehan, Teven Le Scao, Stella Biderman, Leo Gao, Thomas Wolf, and Alexander M. Rush. 2022. Multi-task prompted training enables zero-shot task generalization. In <i>ICLR</i> . OpenReview.net.	795 796 797 798 799 800 801 802 803 804 805 806 807 808 809
	Apoorv Saxena, Aditay Tripathi, and Partha P. Talukdar. 2020. Improving multi-hop question answering over knowledge graphs using knowledge base embeddings. In <i>ACL</i> , pages 4498–4507. Association for Computational Linguistics.	810 811 812 813 814
	Priyanka Sen, Sandeep Mavadia, and Amir Saffari. 2023. Knowledge graph-augmented language models for complex question answering . In <i>Proceedings of the 1st Workshop on Natural Language Reasoning and Structured Explanations (NLRSE)</i> , pages 1–8, Toronto, Canada. Association for Computational Linguistics.	815 816 817 818 819 820 821
	Yu Su, Huan Sun, Brian M. Sadler, Mudhakar Srivatsa, Izzeddin Gur, Zenghui Yan, and Xifeng Yan. 2016. On generating characteristic-rich question sets for QA evaluation. In <i>EMNLP</i> , pages 562–572. The Association for Computational Linguistics.	822 823 824 825 826
	Yiming Tan, Dehai Min, Yu Li, Wenbo Li, Nan Hu, Yongrui Chen, and Guilin Qi. 2023. Evaluation of	827 828

829	chatgpt as a question answering system for answering	882
830	complex questions. <i>CoRR</i> , abs/2303.07992.	883
831	Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier	884
832	Martinet, Marie-Anne Lachaux, Timothée Lacroix,	885
833	Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal	
834	Azhar, Aurélien Rodriguez, Armand Joulin, Edouard	886
835	Grave, and Guillaume Lample. 2023a. Llama: Open	887
836	and efficient foundation language models. <i>CoRR</i> ,	888
837	abs/2302.13971.	889
838	Hugo Touvron, Louis Martin, Kevin Stone, Peter Al-	890
839	bert, Amjad Almahairi, Yasmine Babaei, Nikolay	891
840	Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti	
841	Bhosale, et al. 2023b. Llama 2: Open founda-	892
842	tion and fine-tuned chat models. <i>arXiv preprint</i>	893
843	<i>arXiv:2307.09288</i> .	894
844	Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot,	895
845	and Ashish Sabharwal. 2023. Interleaving retrieval	896
846	with chain-of-thought reasoning for knowledge-	897
847	intensive multi-step questions. In <i>ACL (1)</i> , pages	898
848	10014–10037. Association for Computational Lin-	899
849	guistics.	
850	Chaojie Wang, Yishi Xu, Zhong Peng, Chenxi Zhang,	900
851	Bo Chen, Xinrun Wang, Lei Feng, and Bo An. 2024.	901
852	keqing: knowledge-based question answering is a	902
853	nature chain-of-thought mentor of LLM. <i>CoRR</i> ,	903
854	abs/2401.00426.	904
855	Keheng Wang, Feiyu Duan, Sirui Wang, Peiguang Li,	
856	Yunsen Xian, Chuantao Yin, Wenge Rong, and Zhang	905
857	Xiong. 2023a. Knowledge-driven cot: Exploring	906
858	faithful reasoning in llms for knowledge-intensive	907
859	question answering. <i>CoRR</i> , abs/2308.13259.	908
860	Yufei Wang, Wanjun Zhong, Liangyou Li, Fei Mi,	909
861	Xingshan Zeng, Wenyong Huang, Lifeng Shang,	910
862	Xin Jiang, and Qun Liu. 2023b. Aligning large	911
863	language models with human: A survey. <i>CoRR</i> ,	912
864	abs/2307.12966.	913
865	Jason Wei, Maarten Bosma, Vincent Y. Zhao, Kelvin	914
866	Guu, Adams Wei Yu, Brian Lester, Nan Du, An-	915
867	drew M. Dai, and Quoc V. Le. 2022a. Finetuned	916
868	language models are zero-shot learners. In <i>ICLR</i> .	917
869	OpenReview.net.	918
870	Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten	919
871	Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le,	920
872	and Denny Zhou. 2022b. Chain-of-thought prompt-	921
873	ing elicits reasoning in large language models. In	922
874	<i>NeurIPS</i> .	923
875	Xiaodong Wu, Ran Duan, and Jianbing Ni. 2023a. Un-	924
876	veiling security, privacy, and ethical concerns of chat-	925
877	gpt. <i>CoRR</i> , abs/2307.14192.	926
878	Yike Wu, Nan Hu, Sheng Bi, Guilin Qi, Jie Ren, Anhuan	927
879	Xie, and Wei Song. 2023b. Retrieve-rewrite-answer:	928
880	A kg-to-text enhanced llms framework for knowledge	929
881	graph question answering. <i>CoRR</i> , abs/2309.11206.	930
	Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak	931
	Shafran, Karthik R. Narasimhan, and Yuan Cao. 2023.	932
	React: Synergizing reasoning and acting in language	933
	models. In <i>ICLR</i> . OpenReview.net.	
	Donghan Yu, Sheng Zhang, Patrick Ng, Henghui	
	Zhu, Alexander Hanbo Li, Jun Wang, Yiqun Hu,	
	William Yang Wang, Zhiguo Wang, and Bing Xiang.	
	2023. Decaf: Joint decoding of answers and logical	
	forms for question answering over knowledge bases.	
	In <i>ICLR</i> . OpenReview.net.	
	Susan Zhang, Stephen Roller, Naman Goyal, Mikel	
	Artetxe, Moya Chen, Shuohui Chen, Christopher	
	Dewan, Mona T. Diab, Xian Li, Xi Victoria Lin,	
	Todor Mihaylov, Myle Ott, Sam Shleifer, Kurt Shus-	
	ter, Daniel Simig, Punit Singh Koura, Anjali Srid-	
	har, Tianlu Wang, and Luke Zettlemoyer. 2022.	
	OPT: open pre-trained transformer language mod-	
	els. <i>CoRR</i> , abs/2205.01068.	
	Yichi Zhang, Zhuo Chen, Yin Fang, Lei Cheng, Yanxi	
	Lu, Fangming Li, Wen Zhang, and Huajun Chen.	
	2023. Knowledgeable preference alignment for	
	llms in domain-specific question answering. <i>CoRR</i> ,	
	abs/2311.06503.	
	A Experimental Details	
	A.1 Data Construction Details	
	To construct the supervised fine-tuning dataset, we	
	set the temperature to 0 and adopt GPT-3.5 Turbo.	
	We concatenate question q and its related subgraph	
	G' using a prompt template to form the input x .	
	ChatGPT generates candidate knowledge represen-	
	tation k based on 3 examples as demonstrations	
	and input x under ICL paradigm. Given that, the	
	objective of k is to augment the performance of	
	the QA model, we evaluate the quality of k by ex-	
	amining the QA model’s results. We utilize k as	
	contextual knowledge for the QA model to gener-	
	ate the answer a . If the answer a encompasses	
	all the answer entities, knowledge k is considered	
	helpful for answering the question, and (x, k) is	
	used as an input-output pair for supervised training.	
	For the construction of the preference dataset, we	
	sample knowledge representations from the knowl-	
	edge rewriter after supervised fine-tuning. We set	
	the temperature to 1 to foster greater diversity. Dur-	
	ing preference annotation, given that the knowledge	
	rewriter aims to generate contextual knowledge	
	beneficial for QA, we first assess the quality of	
	knowledge representation based on the number of	
	answer entities they contain. This naïve strategy is	
	robust and cost-effective, avoiding additional API	
	calls for evaluations using ChatGPT, thus saving	
	time and reducing costs. We label the candidate	

with the highest number of answer entities as the preferred knowledge representation k^+ and the one with the fewest as the dispreferred k^- . If the number of answer entities is the same, we select the two, k_1 and k_2 , with the greatest semantic difference by using all-MiniLM-L6-v2³ as the encoder. This selection process ensures a significant semantic gap between the two chosen representations, facilitating more rapid model convergence during training. Subsequently, k_1 and k_2 serve as contextual knowledge to prompt the QA model, yielding answers a_1 and a_2 . We annotate the preferred knowledge representation k^+ and the dispreferred knowledge representation k^- by evaluating the quality of a_1 and a_2 using ChatGPT. Finally, ChatGPT is used to paraphrase the preferred knowledge representation k^+ into an enhanced version k^{++} , forming a preference pair k^{++} and k^- for direct preference optimization (DPO).

A.2 Datasets

GrailQA⁴ (Gu et al., 2021) is a challenging, large-scale multi-hop KGQA benchmark. It is an English dataset that utilizes Freebase (Bollacker et al., 2007, 2008) as KG. It spans 86 domains, such as Sports, Location, and Computer Video Games, and comprises 64,331 questions (44,337 train, 6,763 dev, 13,231 test). This dataset features a large number of entities and relations, complex logical forms, and noise in entity mentions within the questions. The training and dev sets provide annotated SPARQL queries and answer entities, while the test set comprises only the questions. For evaluation convenience, the dev set is used for testing.

GraphQuestions⁵ (Su et al., 2016) is a characteristic-rich dataset for factoid question answering based on Freebase across 70 domains, like People, Astronomy, and Medicine. This English dataset focuses on the following question characteristics: structure complexity, function, commonness, paraphrasing, and answer cardinality. It comprises 5,166 questions (2,771 train, 2,395 test), with nearly half requiring multi-hop reasoning. For each question, the dataset provides corresponding SPARQL query and answer entities.

³<https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>

⁴This dataset is distributed under the CC BY-SA 4.0 license and our utilization complies with the terms specified in the license.

⁵This dataset is licensed under the Creative Commons Attribution 4.0 and our usage aligns with the intended purposes outlined in this license.

A.3 Large Language Models

Llama-2⁶ (Touvron et al., 2023b), an updated version of Llama-1 (Touvron et al., 2023a), is developed using a training corpus comprising 2 trillion tokens and features a context length twice that of Llama-1. To better accomplish the knowledge rewriting task, we select Llama-2-7B-Chat⁷.

Llama-3⁸ (AI@Meta, 2024) is the latest model in the Llama series. It is renowned for its mastery of language nuances, contextual comprehension, and proficiency in executing complex tasks such as translation and generating dialogues. We choose Llama-3-8B-Instruct⁹ for knowledge rewriting.

Mistral¹⁰ (Jiang et al., 2023a) is an open-source LLM developed by Mistral AI. We select the latest instruction-tuned version, Mistral-7B-Instruct-v0.3¹¹, as our QA model.

ChatGPT¹², developed by OpenAI, is a milestone in the era of LLMs. Its robust capabilities in natural language understanding and generation facilitate superior performance across various tasks. We leverage GPT-3.5-turbo via the API¹³ for knowledge rewriting and question answering.

All the LLMs above are general-domain models. Regarding language support, Llama-2, Llama-3, and Mistral only support English, while ChatGPT is multilingual. In this study, the use of these LLMs complies with their respective licenses or terms.

A.4 Retrieval Methods Details

2-Hop subgraph is a naïve question-related context. Most KBQA studies under RAG paradigms consider triples within the N-hop subgraph of the head entity as contextual knowledge (Baek et al., 2023; Sen et al., 2023; Wang et al., 2024; Ko et al., 2024). To retrieve the 2-hop subgraph around the head entity, we execute SPARQL queries on Freebase. Given the large size of the 2-hop subgraph, we

⁶The license of Llama-2 is available at <https://ai.meta.com/resources/models-and-libraries/llama-downloads/>.

⁷<https://huggingface.co/meta-llama/Llama-2-7b-chat-hf>

⁸The license of Llama-3 is available at <https://llama.meta.com/llama3/license>.

⁹<https://huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct>

¹⁰It is under the Apache 2 License.

¹¹<https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.3>

¹²The terms of use for ChatGPT are available at <https://openai.com/policies/terms-of-use/>.

¹³<https://api.openai.com/>

use all-MiniLM-L6-v2¹⁴ to encode all 1-hop and 2-hop relations of the head entity and the question, excluding meaningless relations, such as “common.topic.webpage”. Then, we select semantically similar relation paths based on cosine similarity. Finally, we sample the corresponding triples from KG based on these relation paths. In this experiment, we select the top 30 triples as our retrieval results. However, the small size of the 2-hop subgraphs for some entities may result in fewer than 30 triples being retrieved.

BM25 (Robertson and Zaragoza, 2009) is a retrieval method based on TF-IDF scores of sparse word matching between input questions and passages. This method is commonly used for integrating multimodal data sources or text-based QA (Wang et al., 2023a; Yu et al., 2023). We follow the processing method in DecAF (Yu et al., 2023), simply linearizing the 1-hop subgraph of the topic entity as the article. We use the BM25 implemented by Pyserini (Lin et al., 2021) and collect the triples corresponding to the candidate articles as the retrieval result. Specifically, we initially retrieve 10 candidate articles, each containing up to 10 triples. Subsequently, we remove any redundant triples or those containing meaningless relations. Given the limited context length of LLMs, we select the top 30 triples as the context information for question answering. After filtering, the number of triples in the candidate articles may be less than 30, thus resulting in the retrieval subgraphs for some questions containing fewer than 30 triples.

Ground Truth Subgraph (GS) refers to a subgraph consisting of the triples necessary for answering a question. In this experiment, we modify the SPARQL queries provided in the datasets and execute them on Freebase to obtain the ground truth subgraphs. We use this subgraph to represent the results of an ideal retriever, aiming to explore the performance upper bound of different knowledge rewriting strategies for the QA model.

A.5 Implementation Details

We utilize LoRA (Hu et al., 2022) to achieve parameter-efficient fine-tuning. For supervised fine-tuning and DPO, the batch size, learning rate, lora rank, lora alpha, and lora dropout are set to 128, 1e-4, 64, 128, and 0.05, respectively. In supervised fine-tuning, we train for 10 epochs and save the best model based on validation set results. In DPO,

¹⁴<https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>

we observe that more training steps may lead to decreased model performance. Consequently, we train for 1 to 2 epochs on GraphQuestions using approximately 2,000 training samples, and for 5 to 20 steps on GrailQA using 640 to 2560 training samples. During inference, the temperature is set to 0 for ChatGPT and 0.01 for open-source LLMs, ensuring relatively stable output. All the parameter settings mentioned above are the optimal results after multiple trials.

Our implementation utilizes PyTorch¹⁵, Transformers¹⁶, DeepSpeed¹⁷, Datasets¹⁸, PEFT¹⁹, and TRL²⁰. We use Datasets for data preprocessing. Both training and inference are based on PyTorch and Transformers. Supervised fine-tuning and DPO are implemented using PEFT, TRL, and DeepSpeed. Experiments are conducted on 4 NVIDIA A100-SXM4-40GB GPUs, with each training or inference session completed within one day. Due to the high computational costs of LLMs, we conduct each experiment once and then report the results.

B Prompts

In this section, we provide the prompts (omit demonstrations) used in the experiment.

B.1 Prompts for Knowledge Rewriting Methods

Table 4, Table 5, and Table 6 show the prompts for different knowledge rewriting methods.

KG-to-Text Prompt

[Instruction]

Your task is to transform a knowledge graph to a sentence or multiple sentences. The knowledge graph is: {triples}. The sentence is:

Table 4: KG-to-Text Prompt.

B.2 Prompt for Question Answering

Table 7, Table 8, and Table 9 show the prompts for question answering.

B.3 Prompt for Preference Annotation

Table 10 shows the preference annotation prompt.

¹⁵<https://pytorch.org/>

¹⁶<https://huggingface.co/docs/transformers/en/index>

¹⁷<https://www.deepspeed.ai/>

¹⁸<https://huggingface.co/docs/datasets/en/index>

¹⁹<https://huggingface.co/docs/peft/en/index>

²⁰<https://huggingface.co/docs/trl/en/index>

Summary Prompt

[Instruction]

Your task is to summarize the relevant knowledge that is helpful to answer the question from the following triples.

Triples: {triples}

Question: {question}

Knowledge:

Table 5: Summary Prompt.

CoTKR Prompt

[Instruction]

Your task is to summarize the relevant information that is helpful to answer the question from the following triples. Please think step by step and iteratively generate the reasoning chain and the corresponding knowledge.

Triples: {triples}

Question: {question}

Table 6: CoTKR Prompt.

Prompt for Question Answering with Triple/KG-to-Text/Summary Knowledge

[Instruction]

Your task is to answer the question based on the knowledge that might be relevant. Try to use the original words from the given knowledge to answer the question. But if it is not useful, just ignore it and generate your own guess.

Knowledge: {knowledge}

Question: {question}

Answer:

Table 7: Prompt for Question Answering with Triple/KG-to-Text/Summary Knowledge.

Prompt for Question Answering with CoTKR/CoTKR+PA Knowledge

[Instruction]

Your task is to answer the question based on the reasoning chain that might be relevant. Try to use the original words from the given knowledge to answer the question. But if it is not useful, just ignore it and generate your own guess.

Knowledge: {knowledge}

Question: {question}

Answer:

Table 8: Prompt for Question Answering with CoTKR/CoTKR+PA Knowledge.

B.4 Prompt for Data Augmentation

Table 11 shows the data augmentation prompt.

C Case Study

In this section, we compare different knowledge rewriting strategies through an example. As illustrated in Table 12, the knowledge generated by both Triple and KG-to-Text contains excessive redundant information. This redundancy complicates the process for the QA model, making it challenging to extract relevant knowledge. Summary omits crucial information (i.e., Stephen Jay Gould), rendering the QA model almost unable to provide the correct answer. In contrast, CoTKR and CoTKR+PA emphasize the most pertinent knowledge in the first rewriting step, thereby enabling the QA model to provide a concise and accurate answer. Furthermore, after preference alignment, our knowledge rewriter avoids unnecessary inferential steps (i.e., Reason 2), generating comprehensive and semantic coherent contextual knowledge for QA.

D Additional Experimental Results

This section presents all the experimental results of this study. As shown in Table 13, Table 14, and Table 15, CoTKR/CoTKR+PA achieves the best performance in most scenarios. This indicates that CoTKR is effective for both open-source

LLMs after training and closed-source LLMs using ICL. Besides, the results also reveal the robustness of CoTKR, demonstrating its applicability across KGQA systems with various retrieval methods and QA models.

E AI Assistants in Research or Writing

In this research, we primarily utilize ChatGPT for the construction of training data and as the foundational model for knowledge rewriting and QA. For academic writing, ChatGPT is used to correct grammatical errors.

Prompt for Question Answering without Context

Question: {question}

Answer:

Table 9: Prompt for Question Answering without Context.

Preference Annotation Prompt

[Instruction]

Your task is to evaluate the quality of two responses to the question based on predefined criteria. Avoid any position biases and ensure that the order in which the responses were presented does not influence your decision. Do not allow the length of the responses to influence your evaluation. Be as objective as possible.

[Criteria]

For this evaluation, you should primarily consider the following criteria:

Accuracy: The response should contain as many answer entities as possible, and use the original words of the answer entities.

Relevance: The response should be to the point of the question.

Question: {question}

Answer: {answer}

Response A: {response A}

Response B: {response B}

[Evaluation Rule]

Begin your evaluation by comparing the two responses and provide a short explanation. Then output only the single character: "A" if Response A is better, "B" if Response B is better, and "C" for a tie. At the end, repeat just the letter again by itself on a new line.

Table 10: Preference Annotation Prompt.

Data Augmentation Prompt

[Instruction]

You are a knowledge graph summarizer for Question Answering. I will give you "Question", "Triple", "Answer" and "Knowledge". Your task is to paraphrase the original "Knowledge" into a more helpful representation format for Question Answering. The "Paraphrased Knowledge" should contain the original words of all the answer entities.

Question: {question}

Triple: {triples}

Knowledge: {knowledge}

Paraphrased Knowledge:

Table 11: Data Augmentation Prompt.

Question: what is the central focus of the radio program episode talk of the nation: memes ?		
Answer: Stephen Jay Gould		
KR Methods	KR Result	QA Response
Triple	(Talk of the Nation: Memes, radio.radio_program_episode.program, Talk of the Nation) (Talk of the Nation: Memes, radio.radio_program_episode.subjects, Stephen Jay Gould) (Talk of the Nation, broadcast.content.artist, Neal Conan) (Talk of the Nation, broadcast.content.genre, News)...	The central focus of the radio program episode Talk of the Nation: Memes is the discussion of memes, which are cultural ideas, symbols, or practices that spread from person to person within a culture.
KG-to-Text	"Talk of the Nation: Memes" is an episode of the radio program "Talk of the Nation", which covers topics such as Stephen Jay Gould and his views on atheism and agnosticism. The show features editorial comments from ZSI on Talk of the Nation and is hosted by Neal Conan. Stephen Jay Gould , who was born in Bayside, has children named Jesse and Ethan and received education at various institutions. He is known for his views on atheism and agnosticism and is part of the Ashkenazi Jewish ethnic group.	The central focus of the radio program episode "Talk of the Nation: Memes" is the discussion of memes, which are cultural ideas, symbols, or practices that spread rapidly through society. The show may explore how memes shape our culture and influence our beliefs and behaviors.
Summary	The central focus of the radio program episode "Talk of the Nation: Memes" is discussing topics related to memes and their impact on society.	The central focus of the radio program episode "Talk of the Nation: Memes" is discussing topics related to memes and their impact on society.
CoTKR	Reason: I need to know the subject or focus of the radio program episode "Talk of the Nation: Memes". Summarize: The subject of the radio program episode "Talk of the Nation: Memes" is Stephen Jay Gould . Reason: I need to know more about Stephen Jay Gould. Summarize: Stephen Jay Gould was a person with nationality...	The central focus of the radio program episode "Talk of the Nation: Memes" is Stephen Jay Gould .
CoTKR+PA	Reason: I need to know the subject or focus of the radio program episode "Talk of the Nation: Memes". Summarize: The subject of the radio program episode "Talk of the Nation: Memes" is Stephen Jay Gould .	The central focus of the radio program episode "Talk of the Nation: Memes" is Stephen Jay Gould .

Table 12: An example of knowledge rewriting results for different methods. We use Llama-2 and ChatGPT as the Knowledge Rewriting (KR) backbone and the QA model, respectively.

KR LLMs	Methods	GrailQA			GraphQuestions		
		Acc	Recall	EM	Acc	Recall	EM
<i>ChatGPT as QA model</i>							
None	No Knowledge	28.91	22.81	20.14	35.87	25.76	22.09
	Triple	57.76	49.67	44.73	55.03	46.65	41.63
Llama-2	KG-to-Text	54.75	47.35	42.44	49.73	40.00	33.74
	Summary	58.14	51.38	46.38	52.94	44.70	38.41
	CoTKR	58.64	52.33	47.88	51.36	45.20	39.96
	CoTKR+PA	59.25	53.52	49.64	56.78	47.99	42.46
Llama-3	KG-to-Text	55.76	48.41	43.90	52.40	45.06	39.83
	Summary	57.55	51.06	46.80	54.95	46.86	40.75
	CoTKR	58.33	52.55	48.65	53.19	47.23	43.17
	CoTKR+PA	61.51	56.08	52.67	56.37	49.31	45.26
ChatGPT	KG-to-Text	56.32	49.05	44.73	53.53	45.59	41.17
	Summary	58.54	51.81	47.29	55.62	48.93	44.97
	CoTKR	59.87	53.19	49.02	54.28	48.18	44.68
<i>Mistral as QA model</i>							
None	No Knowledge	29.44	23.13	20.30	38.20	26.92	22.13
	Triple	54.47	47.78	43.25	51.32	45.97	41.67
Llama-2	KG-to-Text	49.49	42.91	38.41	44.59	37.98	32.82
	Summary	54.10	47.79	43.15	49.85	42.33	36.45
	CoTKR	56.75	51.10	46.71	50.19	43.73	38.54
	CoTKR+PA	58.15	52.98	49.13	55.07	47.02	41.71
Llama-3	KG-to-Text	50.64	44.32	40.13	49.06	43.04	38.25
	Summary	53.84	47.71	43.49	52.03	44.30	38.50
	CoTKR	56.47	51.33	47.36	52.65	46.48	42.21
	CoTKR+PA	59.31	54.13	50.24	54.82	47.76	43.09
ChatGPT	KG-to-Text	51.04	44.87	40.97	49.14	43.04	38.83
	Summary	54.44	48.16	43.97	52.28	47.10	43.30
	CoTKR	57.28	51.14	47.09	52.82	47.13	43.55

Table 13: The overall results of CoTKR and the baselines on GrailQA and GraphQuestions using 2-Hop as retrieval method. For each combination of the Knowledge Rewriter (KR) LLM and the QA model, the best and second-best results are highlighted in bold and underlined, respectively.

KR LLMs	Methods	GrailQA			GraphQuestions		
		Acc	Recall	EM	Acc	Recall	EM
ChatGPT as QA model							
None	No Knowledge	28.91	22.81	20.14	35.87	25.76	22.09
	Triple	58.42	49.06	43.87	48.43	40.44	36.83
	KG-to-Text	52.80	43.93	39.01	41.67	32.09	27.68
Llama-2	Summary	57.62	49.13	43.80	46.47	36.84	31.44
	CoTKR	58.32	50.32	45.57	45.34	37.21	32.73
	CoTKR+PA	58.32	50.77	46.41	49.52	40.06	35.16
Llama-3	KG-to-Text	55.14	46.64	41.96	44.68	34.51	30.19
	Summary	59.19	50.83	46.19	44.97	36.97	32.44
	CoTKR	58.89	51.13	46.93	46.18	38.59	35.03
ChatGPT	CoTKR+PA	59.69	52.34	48.14	50.44	42.03	37.58
	KG-to-Text	55.61	47.00	42.38	46.14	36.64	31.98
	Summary	58.76	50.47	45.66	46.14	39.20	35.66
	CoTKR	59.66	51.06	46.40	45.18	38.03	34.36
Mistral as QA model							
None	No Knowledge	29.44	23.13	20.30	38.20	26.92	22.13
	Triple	55.58	47.10	42.07	43.34	36.98	33.86
	KG-to-Text	47.33	39.90	35.64	35.24	28.35	25.05
Llama-2	Summary	52.94	44.94	40.12	43.47	34.56	29.52
	CoTKR	55.69	48.37	43.77	45.39	37.07	32.73
	CoTKR+PA	57.13	50.17	45.94	49.02	39.46	34.53
Llama-3	KG-to-Text	49.83	42.40	38.07	39.21	31.86	28.48
	Summary	54.33	46.73	42.35	41.63	34.41	30.15
	CoTKR	56.76	49.75	45.57	44.38	37.14	33.57
ChatGPT	CoTKR+PA	58.27	51.00	46.72	48.14	39.34	34.91
	KG-to-Text	50.35	42.69	38.40	39.46	33.36	29.52
	Summary	54.98	47.15	42.63	41.42	35.60	31.86
	CoTKR	56.84	48.87	44.17	43.42	36.23	32.57

Table 14: The overall results of CoTKR and the baselines on GrailQA and GraphQuestions using BM25 as retrieval method. For each combination of the Knowledge Rewriter (KR) LLM and the QA model, the best and second-best results are highlighted in bold and underlined, respectively.

KR LLMs	Methods	GrailQA			GraphQuestions		
		Acc	Recall	EM	Acc	Recall	EM
ChatGPT as QA model							
None	No Knowledge	28.91	22.81	20.14	35.87	25.76	22.09
	Triple	77.41	67.14	61.47	83.17	70.89	62.76
Llama-2	KG-to-Text	80.05	70.20	65.06	76.24	64.47	57.58
	Summary	85.14	75.90	70.03	80.88	69.31	60.00
	CoTKR	87.18	78.46	73.56	80.67	70.73	63.72
	CoTKR+PA	87.79	79.86	75.50	83.67	75.02	68.85
Llama-3	KG-to-Text	82.17	73.10	68.28	78.66	68.32	59.92
	Summary	85.44	77.01	72.07	83.55	72.47	64.47
	CoTKR	88.27	80.35	75.93	85.51	75.89	69.14
	CoTKR+PA	91.48	84.02	79.93	87.85	79.66	73.95
ChatGPT	KG-to-Text	80.39	71.19	66.46	83.05	73.60	67.22
	Summary	86.53	77.88	72.87	88.39	79.26	72.99
	CoTKR	87.77	78.97	74.38	89.81	80.09	73.49
Mistral as QA model							
None	No Knowledge	29.44	23.13	20.30	38.20	26.92	22.13
	Triple	74.12	65.83	60.79	83.26	72.17	64.22
Llama-2	KG-to-Text	72.47	64.18	59.53	71.19	59.62	51.98
	Summary	78.87	70.43	64.71	78.71	67.45	58.29
	CoTKR	84.27	76.57	71.60	78.66	68.75	61.80
	CoTKR+PA	85.81	79.26	74.52	81.38	72.88	67.06
Llama-3	KG-to-Text	74.54	66.58	62.38	76.37	66.56	58.75
	Summary	79.64	71.65	66.89	80.84	70.07	62.21
	CoTKR	84.73	77.86	73.31	84.18	75.04	68.64
	CoTKR+PA	87.80	81.24	76.65	84.63	77.38	73.24
ChatGPT	KG-to-Text	72.53	65.06	60.92	81.71	72.35	65.43
	Summary	81.34	72.53	67.53	86.47	77.15	70.56
	CoTKR	84.77	76.72	72.11	88.60	78.95	72.36

Table 15: The overall results of CoTKR and the baselines on GrailQA and GraphQuestions using GS as retrieval method. For each combination of the Knowledge Rewriter (KR) LLM and the QA model, the best and second-best results are highlighted in bold and underlined, respectively.